

1. *Linear Regression*

2. *Simple linear regression*

3. *Multiple linear regression*

4. *Importance of coeff*

5. *Error, SSE, MSE, RMSE, RSS, LOSS, COST*

6. *OLS method*

7. *Interpretation*

8. *RSE*

9. *Rsquare*

10. *Adj Rsquare*

11. *AIC*

12. *BIC*

13. *Overfit – underfit*

14. *Bias – variance trade off*

15. *Hypothesis*

16. *Assumptions*

Why overfitting

1. *Less data*

2. *bias – variance x – axis: Model complexity*

- *as model complexity increases, we are adding the more variables*
- *we are adding some unwanted variables*
- *Model become overfit*
- *low bias – high variance*

It is very important to do feature selection

Hypothesis in Linear Regression

Statement is:

There is No relation between dependent variable and independent variable

Null Hypothesis :

H_0 : *There is No relation between dependent variable(y) and independent variable(x)*

Alternative Hypothesis :

H_1 : *There is a Relation between dependent variable(y) and independent variable(x)*

Step – 1: Form the Hypothesis

$$Y = b_0 + b_1 * x_1$$

H_0 : *There is No relation between dependent variable (y) and independent variable(x)*

H_1 : *There is a Relation between dependent variable(x) and independent variable(x)*

Step – 2: Make math representation

$$H_0: b_1 = 0$$

$$H_1: b_1 \neq 0$$

Step – 3: Set significance level

$$\alpha = 0.05$$

Step – 4: Make the rejection and Non rejection area (In house)

Step – 5: z or t test

$$z = \frac{\bar{x} - u}{SE} = \frac{b_{1Sample} - 0}{SE(b_1)}$$

Step – 6: calculate p – value

Step – 7: Conclusion $p < \alpha$: Reject the H_0

$p < \alpha$: *Reject the H_0*

- Null hypothesis is wrong
- Relation is there
- Include that variable

$p > \alpha$: not Reject the H_0

- Null hypothesis correct
- No Relation
- Do not Include that variable

Whenever in the output if you see

p – value is more then do not include that variable

$$\text{Revenue} = b_0 + b_1 * \text{promotion}$$

$$\text{Revenue} = 881382.223 + 2.423 * \text{promotion}$$

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Call:
lm(formula = ROLL ~ UNEM + HGRAD + INC, data = datavar)

Residuals:
    Min       1Q   Median       3Q      Max
-1148.840 -489.712  -1.876   387.400  1425.753

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -9.153e+03  1.053e+03  -8.691 5.02e-09 ***
UNEM         4.501e+02  1.182e+02   3.809 0.000807 ***
HGRAD        4.065e+01  7.602e+02   5.347 1.52e-05 ***
INC          4.275e+00  4.947e-01   8.642 5.59e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 670.4 on 25 degrees of freedom
Multiple R-squared:  0.9621, Adjusted R-squared:  0.9576
F-statistic: 211.5 on 3 and 25 DF, p-value: < 2.2e-16
```

OLS Regression Results						
Dep. Variable:	y	R-squared:	0.669			
Model:	OLS	Adj. R-squared:	0.667			
Method:	Least Squares	F-statistic:	299.2			
Date:	Mon, 01 Mar 2021	Prob (F-statistic):	2.13e-37			
Time:	16:19:34	Log-Likelihood:	-88.666			
No. Observations:	150	AIC:	181.4			
DF Residuals:	148	BIC:	187.4			
DF Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	-3.2082	0.257	-12.458	0.000	-3.708	-2.693
x1	0.7529	0.044	17.296	0.000	0.667	0.839
Omnibus:	3.538	Durbin-Watson:		1.279		
Prob(Omnibus):	0.171	Jarque-Bera (JB):		3.589		
Skew:	0.157	Prob(JB):		0.166		
Kurtosis:	2.744	Cond. No.		43.4		

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call:
lm(formula = sales ~ ., data = adv_training)

Residuals:
    Min       1Q   Median       3Q      Max
-8.6331 -0.8971  0.2283  1.1971  2.9630

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.707724   0.354788   7.632 2.05e-12 ***
TV           0.046521   0.001561  29.801 < 2e-16 ***
radio        0.188857   0.009423  20.041 < 2e-16 ***
newspaper    0.001619   0.006255   0.259  0.796
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
Residual standard error: 1.677 on 158 degrees of freedom
Multiple R-squared:  0.8967,    Adjusted R-squared:  0.894
F-statistic: 457.1 on 3 and 158 DF,  p-value: < 2.2e-16

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The multiple regression output

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

First, we firm perform F test.
F statistic is a test of significance for the entire regression. At $\alpha = 0.05$, this regression is statistically significant because p-value < 0.05.

ANOVA					
	df	SS	MS	F	P-value
Regression	k-1 = 2	1,326.54	663.2683	14.4695	0.0002
Residual/Error	n-k = 17	779.26	45.8390		
Total	n-1 = 19	2,105.80			

	Coeff.	Std. Err.	t Stat	P-value	Lower 95%	Upper 95%
Intercept	-25.6758	16.1804	-1.5868	0.1310	-59.8136	8.4619
X1(distance in miles)	1.2125	0.2595	4.6730	0.0002	0.6651	1.7599
X2 (# items)	-0.5673	0.2310	-2.4556	0.0251	-1.0547	-0.0799

Prediction interval for E(Y)

Confidence interval for β_1

Confidence interval for β_2

Note that k = 3 parameters to be estimated ($\beta_0, \beta_1, \beta_2$)

At $\alpha = 0.05$, both t-values are statistically significant because their corresponding p-values < 0.05. Therefore, both X_1 and X_2 are individually useful in the prediction of Y.

D	E	F	G	H	I	J	K
SUMMARY OUTPUT							
<i>Regression Statistics</i>							
Multiple R	0.857						
R Square	0.734						
Adjusted R Square	0.703						
Standard Error	5.366						
Observations	20						
ANOVA							
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>		
Regression	2	1350.76	675.38	23.46	0.00		
Residual	17	489.44	28.79				
Total	19	1840.20					
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	
Intercept	67.67	2.82	24.03	0.00	61.73	73.61	
hours	5.56	0.90	6.18	0.00	3.66	7.45	
prep_exams	-0.60	0.91	-0.66	0.52	-2.53	1.33	

OLS Regression Results

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Dep. Variable:          sales    R-squared:                0.278
Model:                  OLS      Adj. R-squared:            0.275
Method:                 Least Squares    F-statistic:            127.0
Date:                  Sun, 11 Nov 2018    Prob (F-statistic):      0.00
Time:                  14:48:08    Log-Likelihood:         -27899.
No. Observations:      5967    AIC:                    5.584e+04
Df Residuals:          5948    BIC:                    5.596e+04
Df Model:              18
Covariance Type:       nonrobust
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	coef	std err	t	P> t	[0.025	0.975]
Intercept	7.4966	6.575	1.140	0.254	-5.394	20.387
weekend[T.Y]	12.4705	0.757	16.472	0.000	10.986	13.955
holiday[T.Y]	12.5009	1.911	6.542	0.000	8.755	16.247
prod_function[T.EyeCare]	-17.0896	1.738	-9.835	0.000	-20.496	-13.683
prod_function[T.Moisturisers]	-4.2329	1.081	-3.914	0.000	-6.353	-2.113
prod_function[T.Other]	-1.4569	1.600	-0.911	0.363	-4.594	1.680
prod_function[T.Toner]	-8.2722	1.626	-5.089	0.000	-11.459	-5.085
prod_function[T.Treatments]	-9.7475	1.246	-7.820	0.000	-12.191	-7.304
prod_regime[T.HERB]	14.5523	1.942	7.495	0.000	10.746	18.359
prod_regime[T.OCEAN]	1.0748	1.343	0.800	0.424	-1.558	3.708
prod_regime[T.POME]	1.7154	1.896	0.905	0.366	-2.002	5.433
prod_regime[T.TEA]	16.3640	1.141	14.337	0.000	14.127	18.601
prod_regime[T.VITA]	6.6399	1.106	6.003	0.000	4.472	8.808
prod_regime[T.WHITE]	23.1618	2.031	11.405	0.000	19.181	27.143
prod_regime[T.YOUTH]	47.4428	2.708	17.518	0.000	42.134	52.752
month	2.5016	0.683	3.660	0.000	1.162	3.841
prod_listprice	-7.49e-05	7.77e-06	-9.642	0.000	-9.01e-05	-5.97e-05
discount	0.4146	0.035	12.005	0.000	0.347	0.482
stockout	-32.6950	1.124	-29.086	0.000	-34.899	-30.491

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Omnibus:                5081.500    Durbin-Watson:           1.821
Prob(Omnibus):          0.000    Jarque-Bera (JB):       201849.324
Skew:                   3.904    Prob(JB):                0.00
Kurtosis:               30.403    Cond. No.:               4.89e+06
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